

Representative AI-Assisted Development Requests

Streaming Event Anomaly Detection Platform · 12 Development Areas · Generated June 2026

During development of the Streaming Event Anomaly Detector project, AI tools were leveraged through structured technical requests across twelve core development areas. Each section below documents the nature of the AI assistance requested, providing a transparent record of AI-assisted design decisions, implementation strategies, and architectural guidance used throughout the project lifecycle.

AREAS OF AI ASSISTANCE

01	System Architecture Design	02	Statistical Anomaly Detection	03	Event Streaming Simulation
04	AI-Powered Analysis	05	Agent-Based Decision Support	06	Database Integration
07	API Development	08	Authentication and Authorization	09	Dashboard Development
10	Service Integration	11	Containerization and Deployment	12	Testing and Validation

DETAILED REQUEST DESCRIPTIONS

01

System Architecture Design

Architecture

Requested guidance on designing an end-to-end streaming event anomaly detection platform, including event generation, anomaly detection, AI analysis, storage, and visualization components.

02

Statistical Anomaly Detection

Analytics

Requested implementation strategies for real-time anomaly detection using sliding-window processing and Z-Score based statistical analysis.

03

Event Streaming Simulation

Simulation

Requested approaches for simulating realistic streaming order events, including normal traffic patterns, traffic spikes, drops, and burst scenarios.

04

AI-Powered Analysis

AI / LLM

Requested methods for integrating Large Language Models to generate human-readable explanations, root-cause insights, and anomaly interpretations.

05

Agent-Based Decision Support

AI Agent

Requested implementation guidance for an AI Agent capable of classifying anomaly severity, generating recommendations, and supporting operational decision-making.

06

Database Integration

Storage

Requested assistance in designing SQLite schemas, persistence mechanisms, event storage workflows, and anomaly history management.

07

API Development

REST API

Requested support for building RESTful APIs to expose metrics, events, anomalies, authentication services, and dashboard data.

08

Authentication and Authorization

Security

Requested implementation guidance for secure login mechanisms, role-based access control, JWT authentication, and permission management.

09

Dashboard Development

Frontend

Requested recommendations for building an interactive monitoring dashboard capable of visualizing live metrics, anomaly history, AI insights, and system status.

10

Service Integration

Notifications

Requested integration patterns for external notification services, alerting workflows, and operational communication channels.

11

Containerization and Deployment

DevOps

Requested assistance for creating Docker-based deployment configurations to ensure environment consistency across development and demonstration systems.

12

Testing and Validation

QA

Requested approaches for validating anomaly detection logic, verifying API behavior, and creating test scenarios covering expected operational conditions.